



Data warehousing and Business Intelligence - IT3021

IT23720756 – Abilas M A S

Assignment 2

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Data Source

This dataset captures the operational transactions of a pharmacy, including medical sales, inventory management, and staff interactions.

It is modeled after an OLTP (Online Transaction Processing) system that records daily activities such as prescription fulfillment, retail sales, and staff assignments.

Entities and Their Purpose

1. Dim_Product

- Stores the product catalog and its hierarchical classification.
- Attributes: ProductKey , ProductID , Category , Subcategory , Manufacturer

2. Dim_Pharmacy

- Contains details regarding the physical branch locations where sales occur.
- Attributes: PharmacyKey , PharmacyID , City, Region

3. Dim_Staff

- Tracks professional details of staff members and their roles. It is implemented as a Slowly Changing Dimension (SCD Type 2) to maintain historical accuracy of roles.
- **Attributes:** StaffKey , StaffID , StaffName, Role, StartDate, EndDate, IsCurrent.

4. Fact_Sales

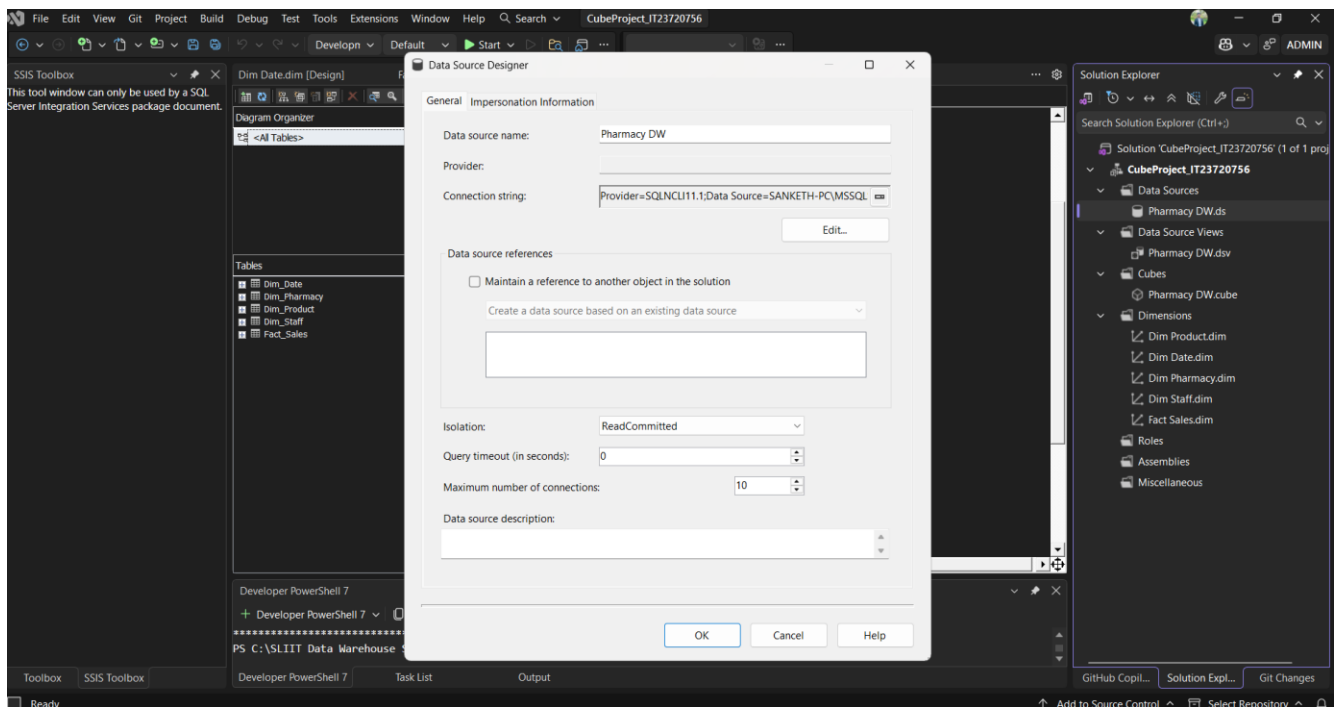
- The central fact table representing individual sales transactions. It is designed as an Accumulating Snapshot Fact Table to track the processing lifecycle of a transaction.
- **Purpose:** Stores numerical measures for sales performance and calculates processing efficiency
- **Attributes:** TransactionID, ProductKey, PharmacyKey, StaffKey, DateKey, QuantityPack, UnitPrice, TotalAmount, AccmTxnCreateTime, AccmTxnCompleteTime

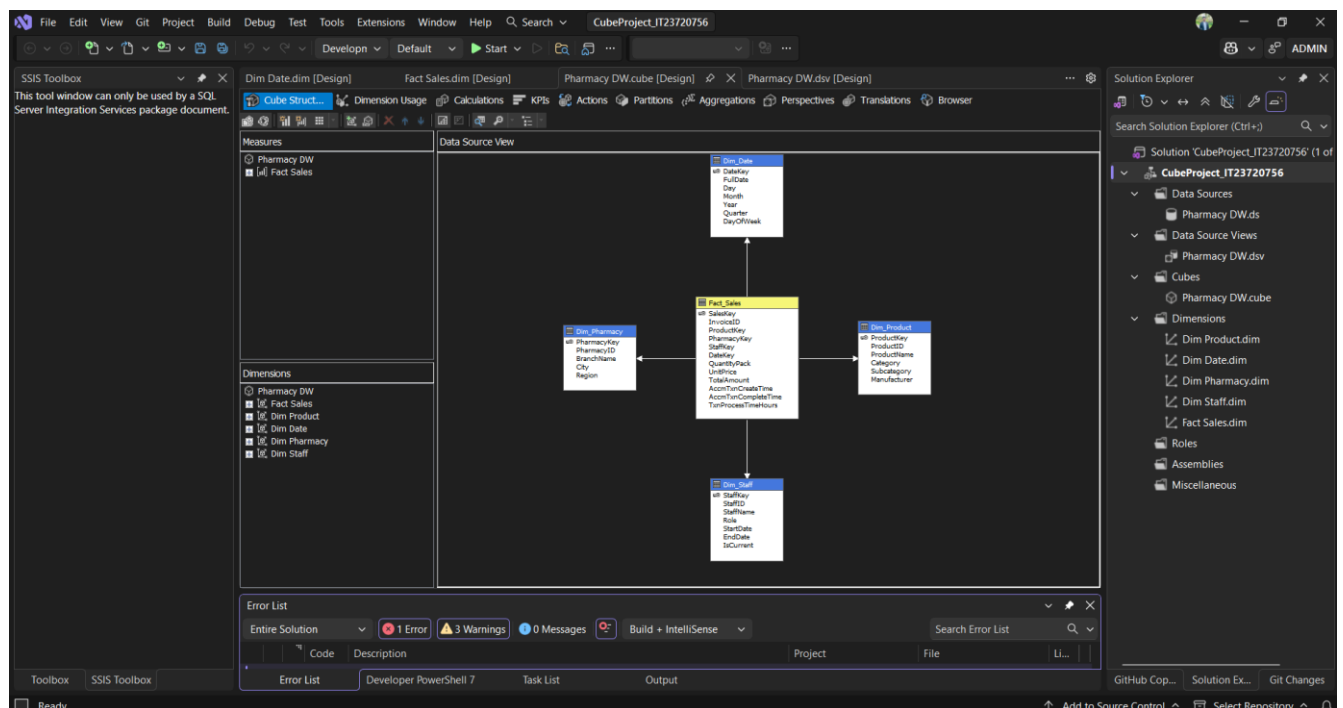
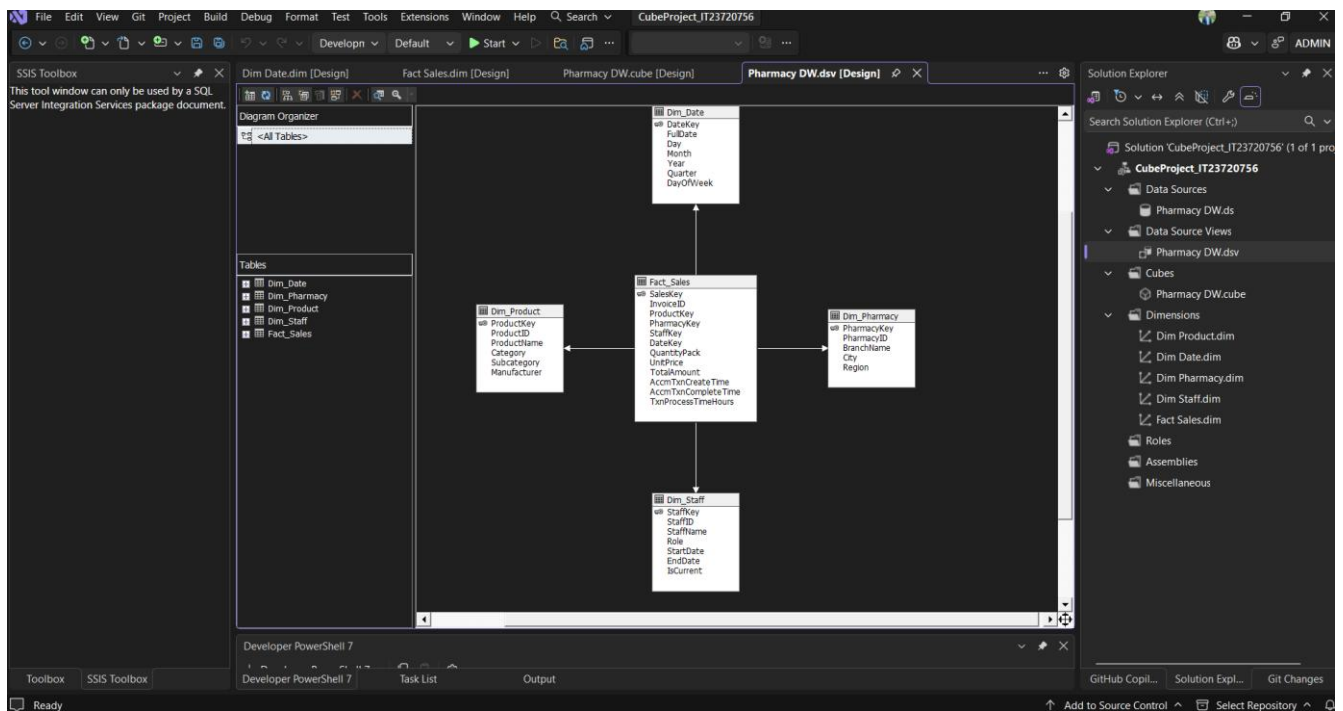
5. DimDate

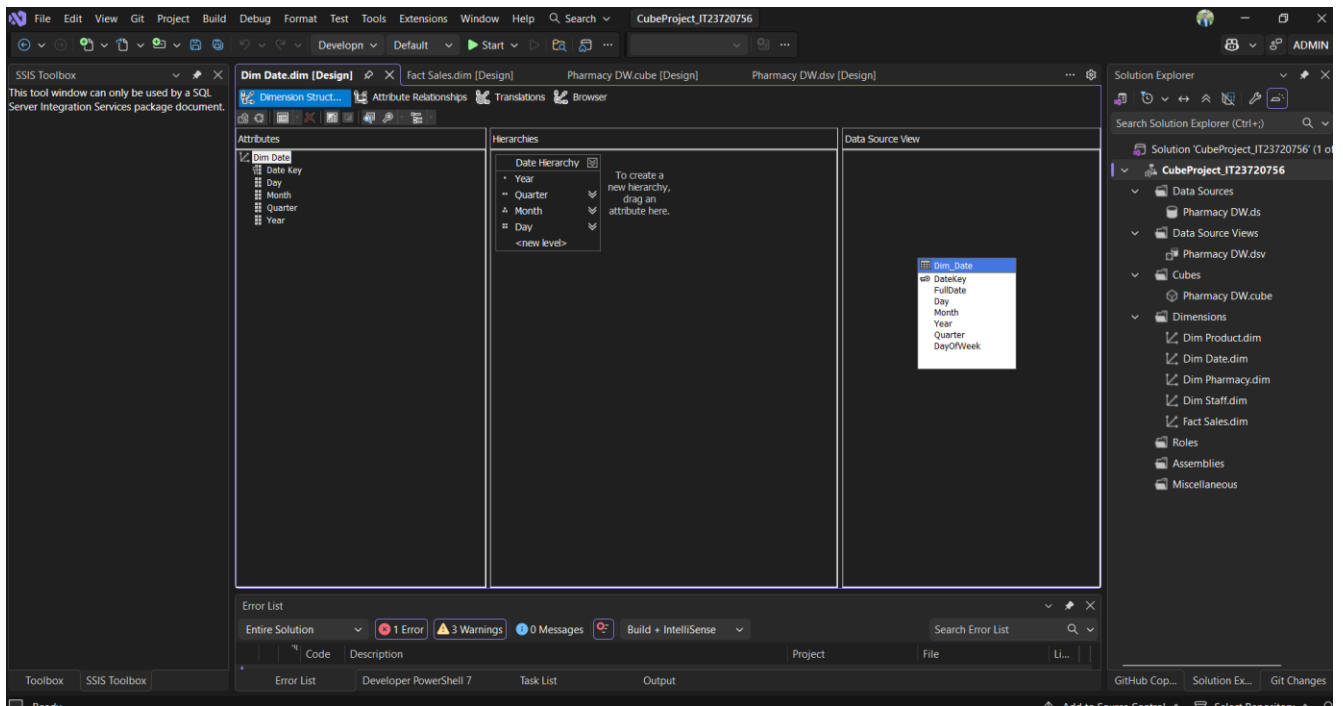
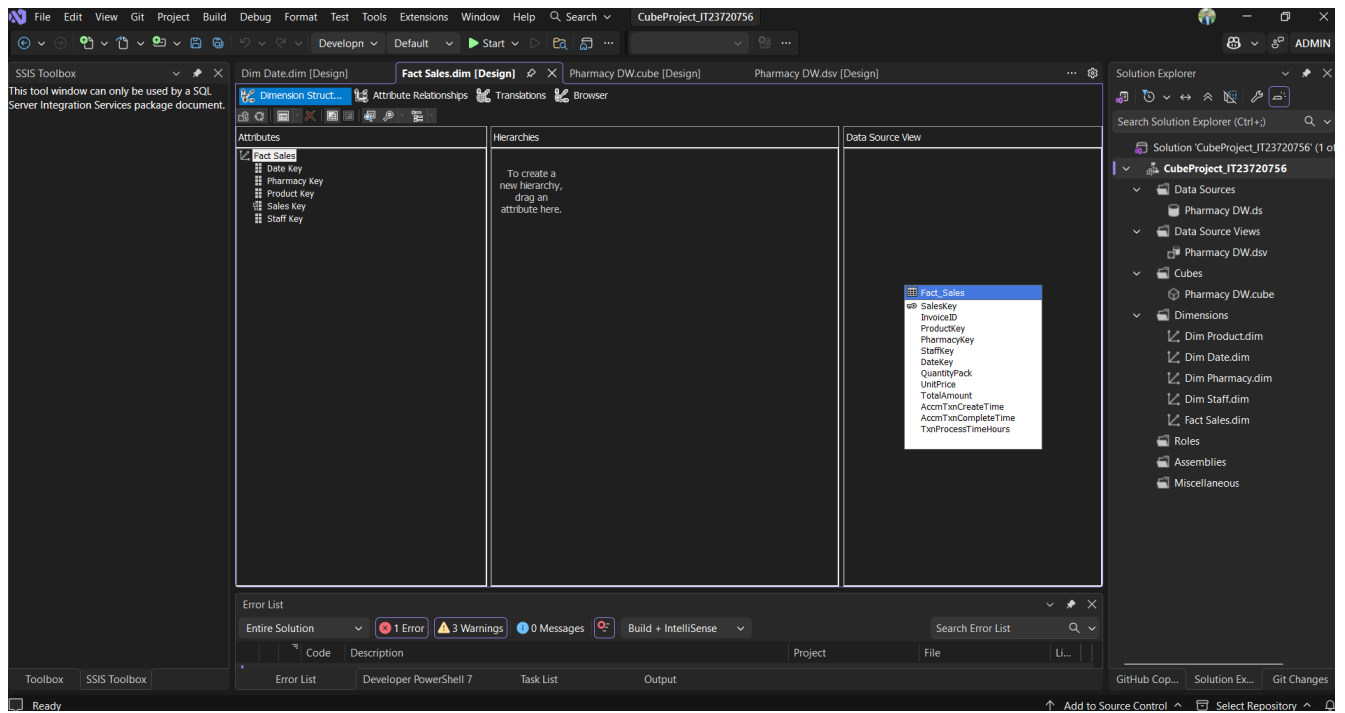
- Includes the dates where the transaction are done.

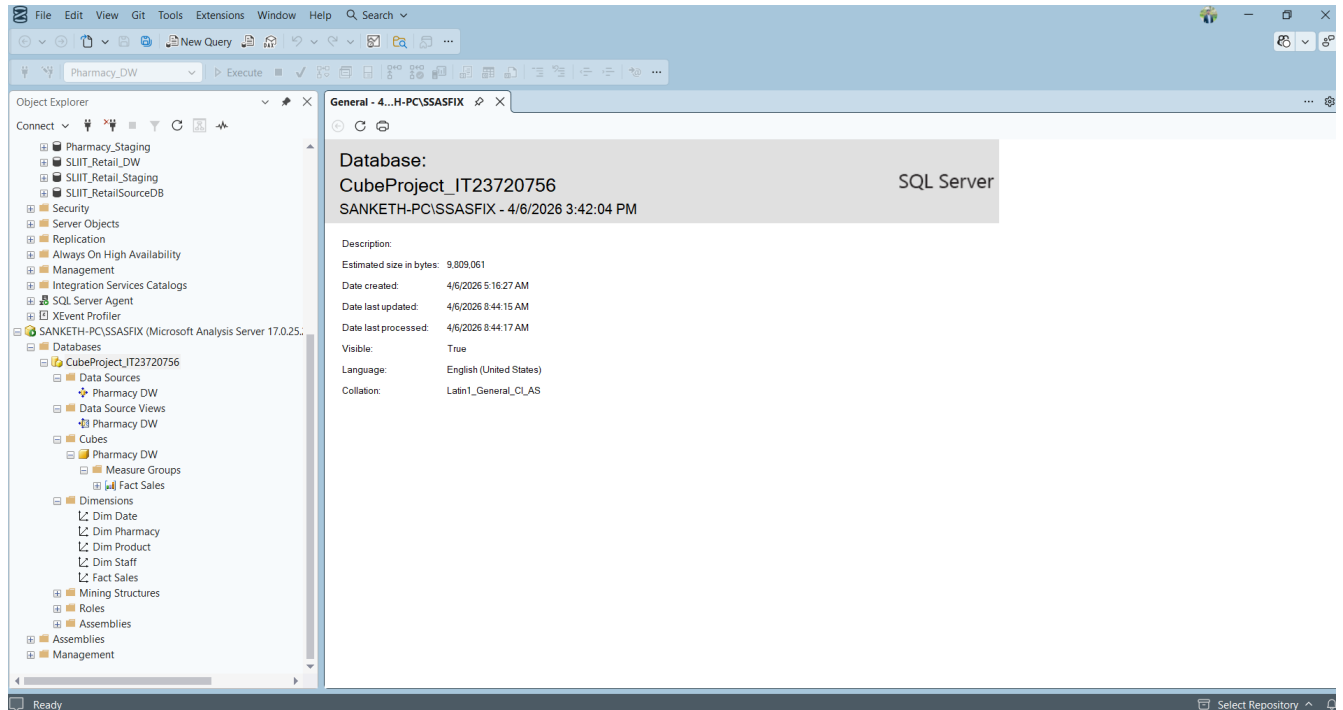
SSAS Cube Implementation

- First, I loaded my data in the data warehouse to the Data Sources tab
- Next, I created the Data ware house view using data warehouse.
- Then I started to use the implement the cube by using the wizard. Mentioned the relevant fact tables and dimension tables as well.



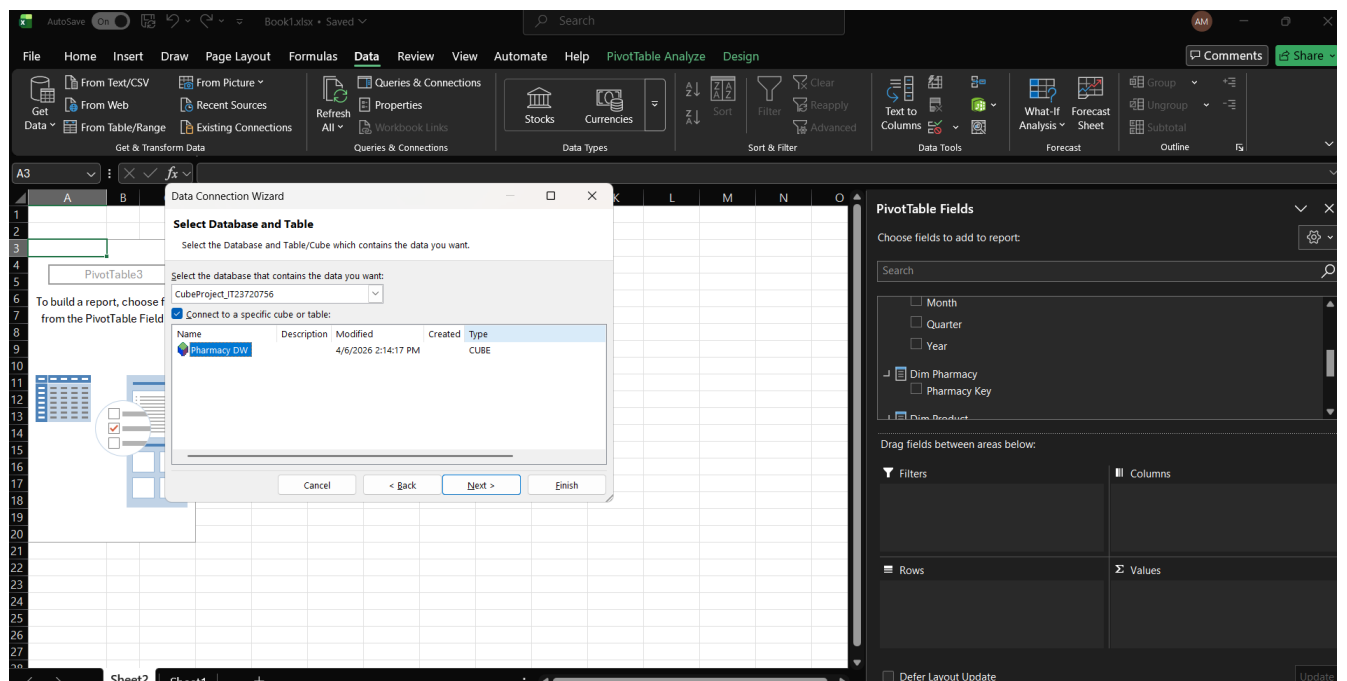






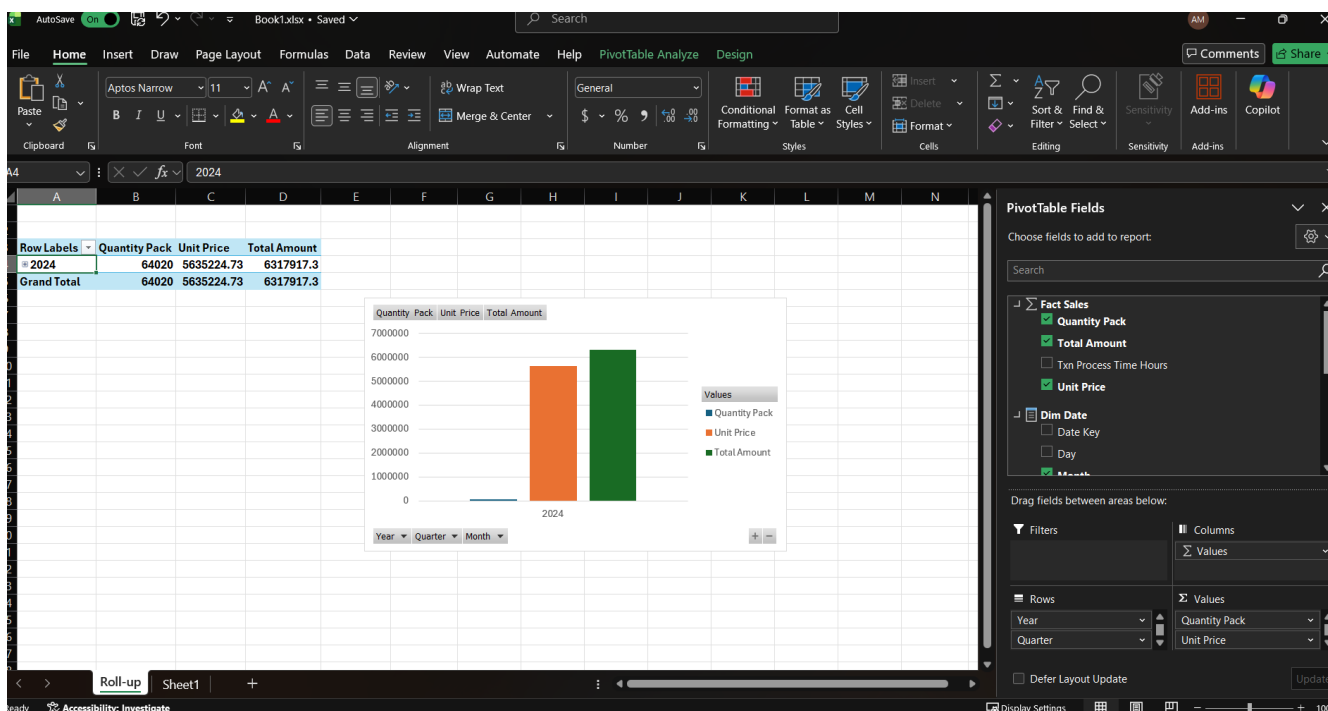
OLAP operations

- After loading the data into the cube I use the excel to create the pivot tables using the cube.



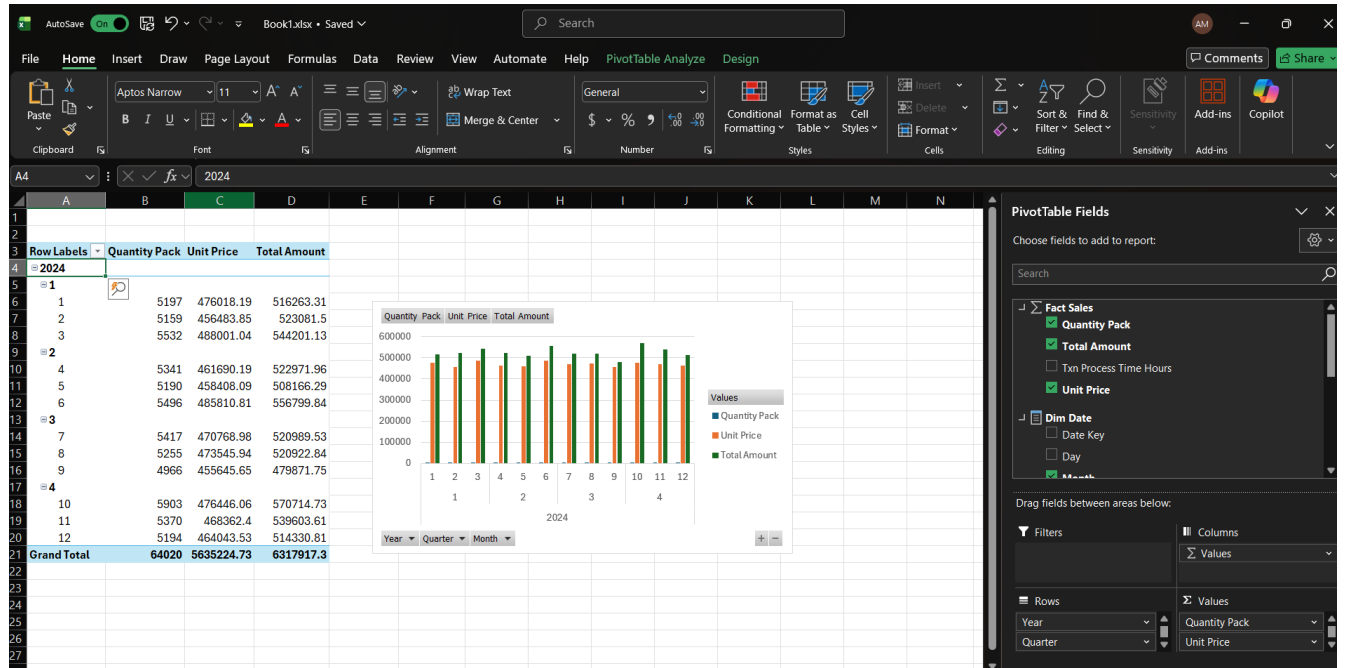
1. Roll-Up

- I clicked the minus (-) button on the "2024" Year and "Quarter" labels to collapse the months.



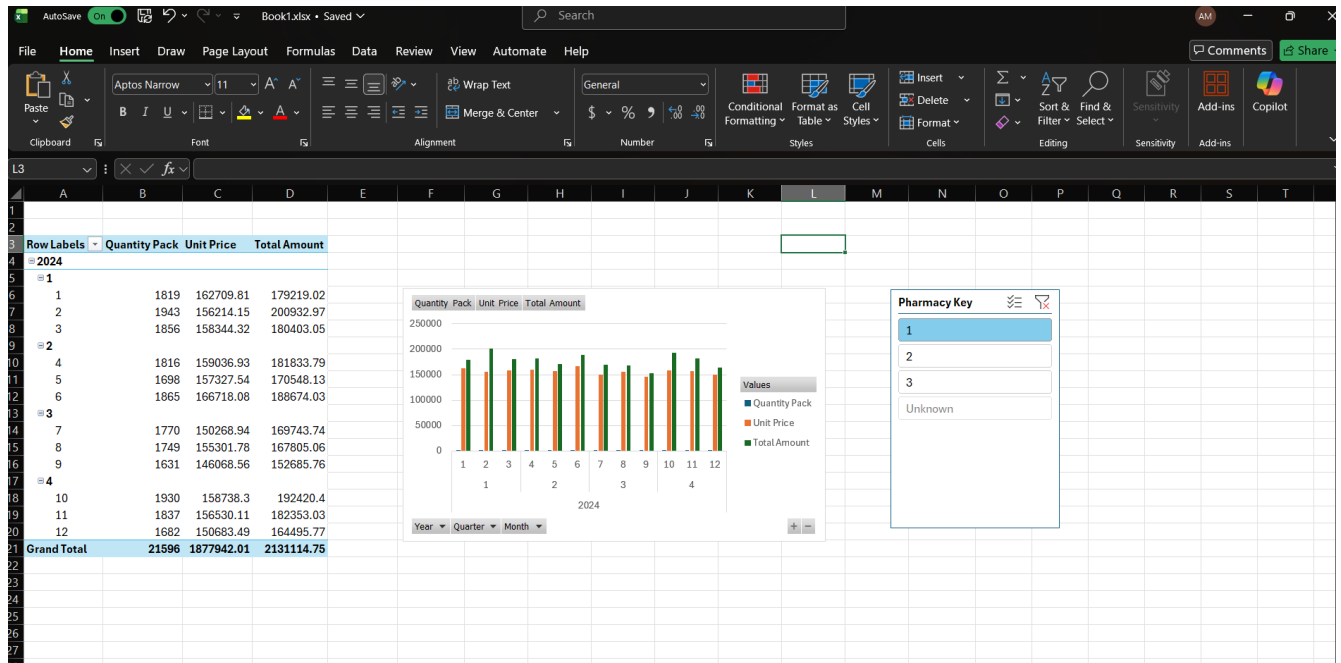
2. Drill-Down

- I clicked the plus (+) button next to the "2024" year label.



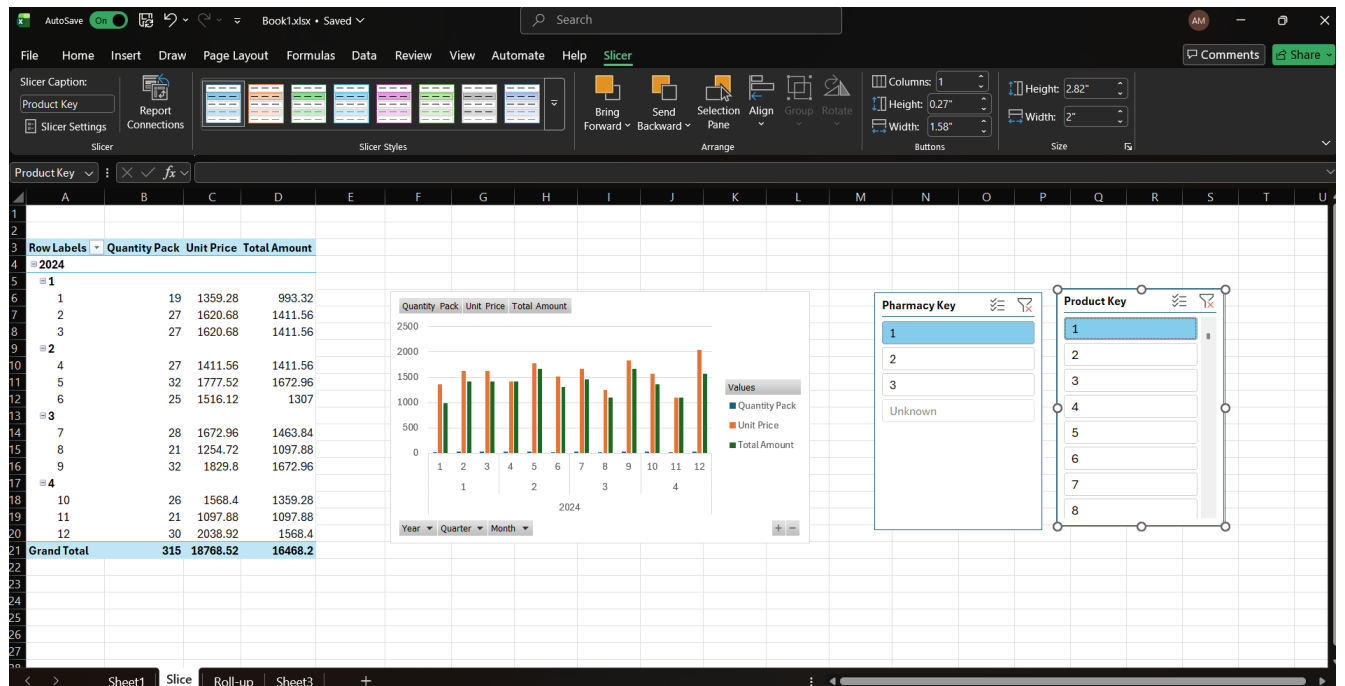
3. Slice

- I inserted a Slicer for Pharmacy Key and selected a single pharmacy ID.



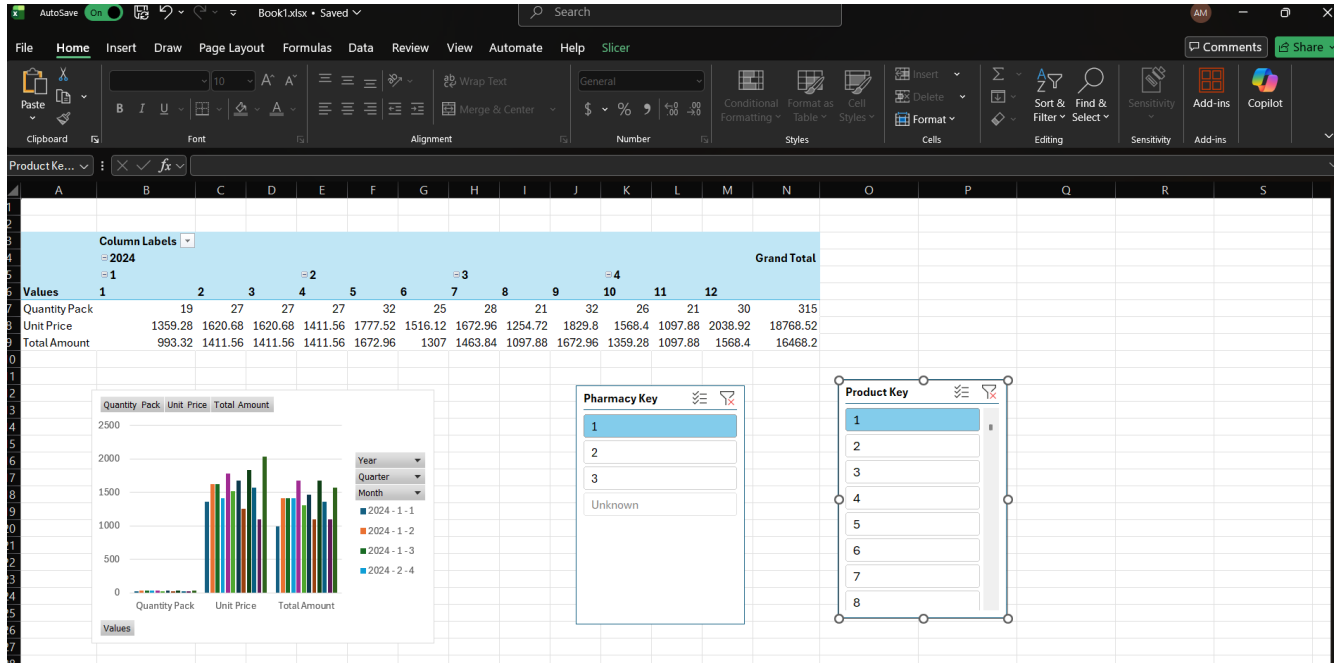
4. Dice

- I added a second Slicer for Product Key alongside the Pharmacy Key slicer.



5. Pivot

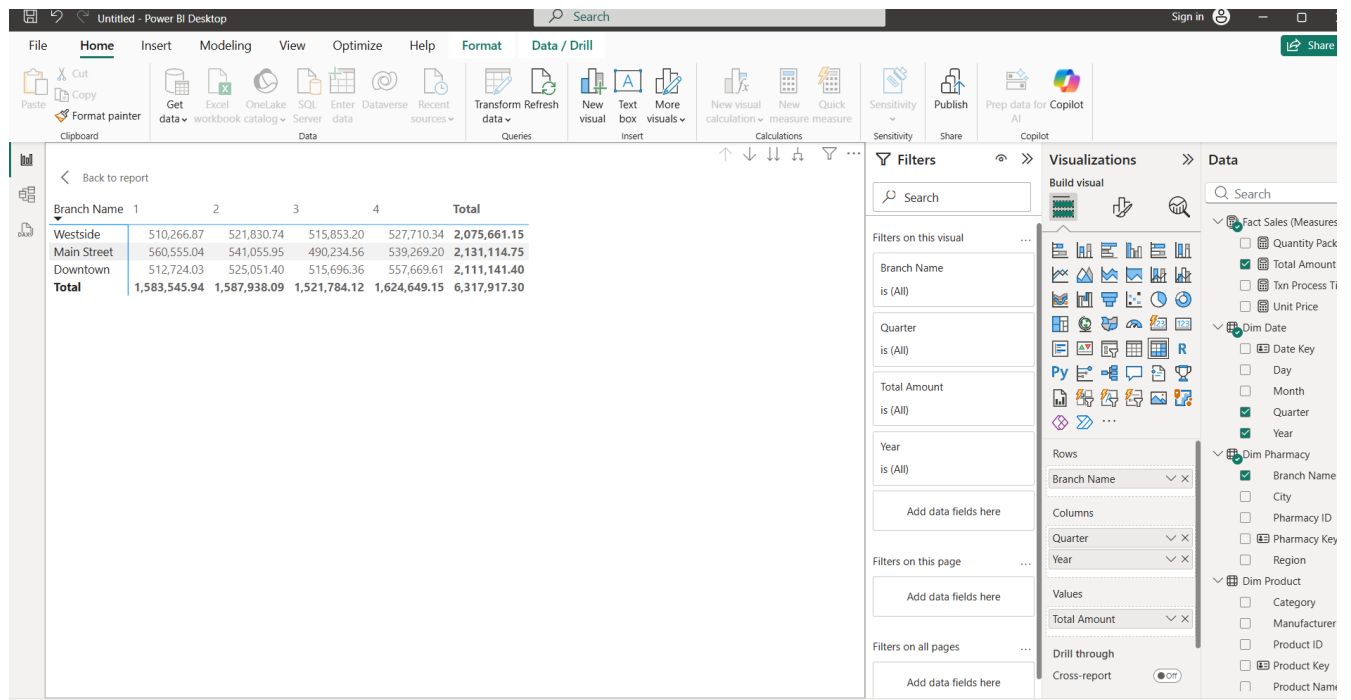
- I dragged the Year and Quarter fields from the Rows area into the Columns area in the PivotTable Fields pane.



PowerBI Reports

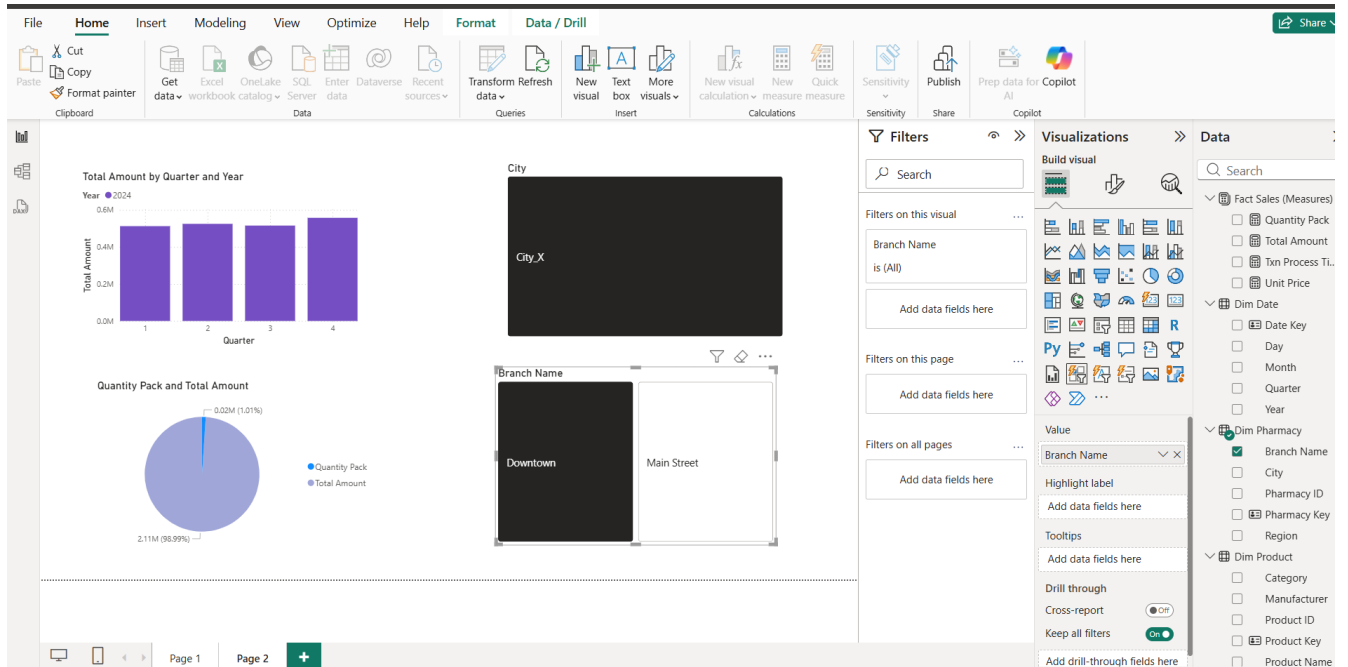
1. Report 1

- Selected the Matrix visual. Dragged Branch Name to Rows, Year and Quarter to Columns, and Total Amount to Values. This created a hierarchical grid with automatic subtotals.



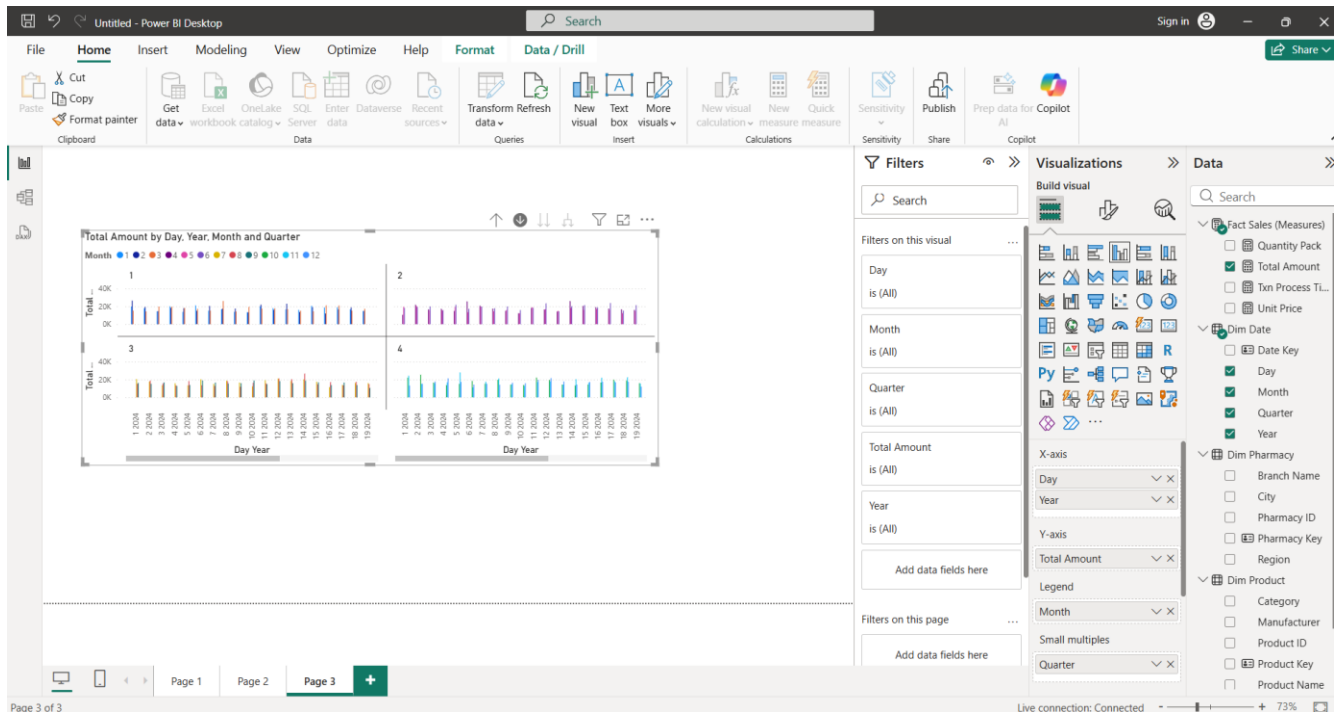
2. Report 2

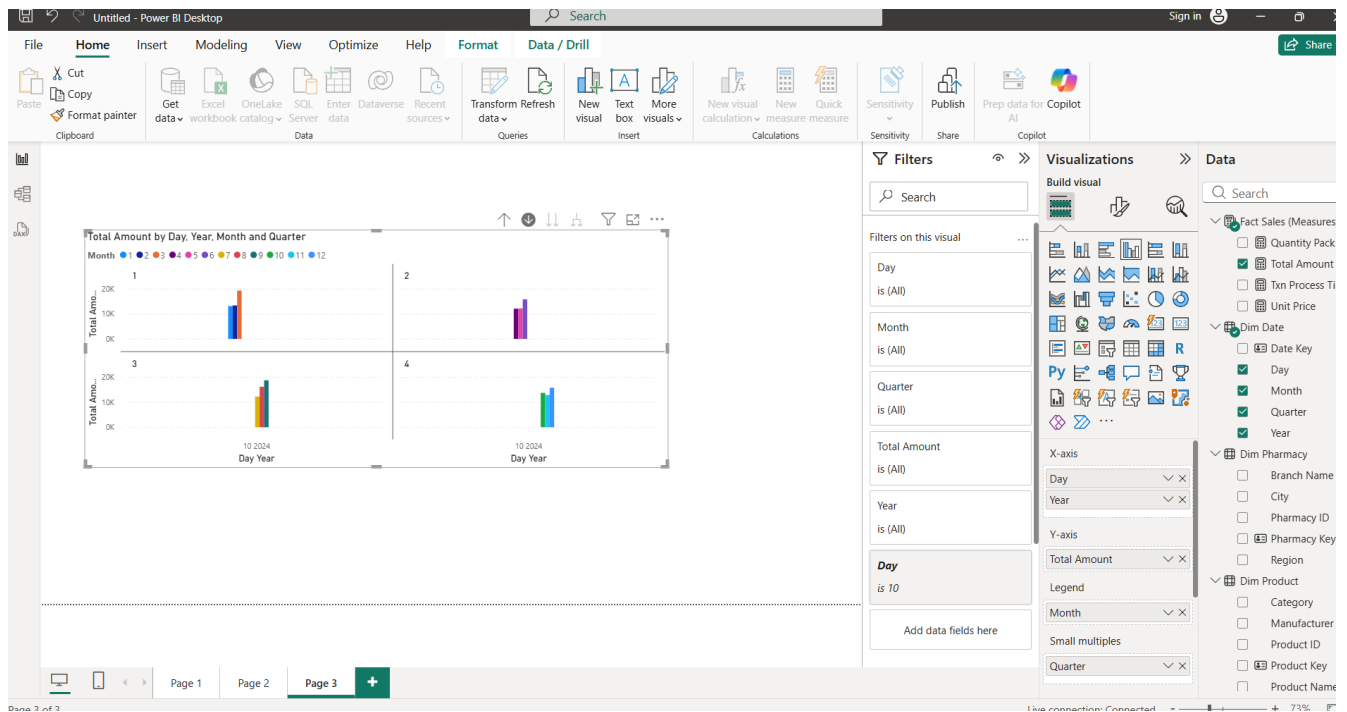
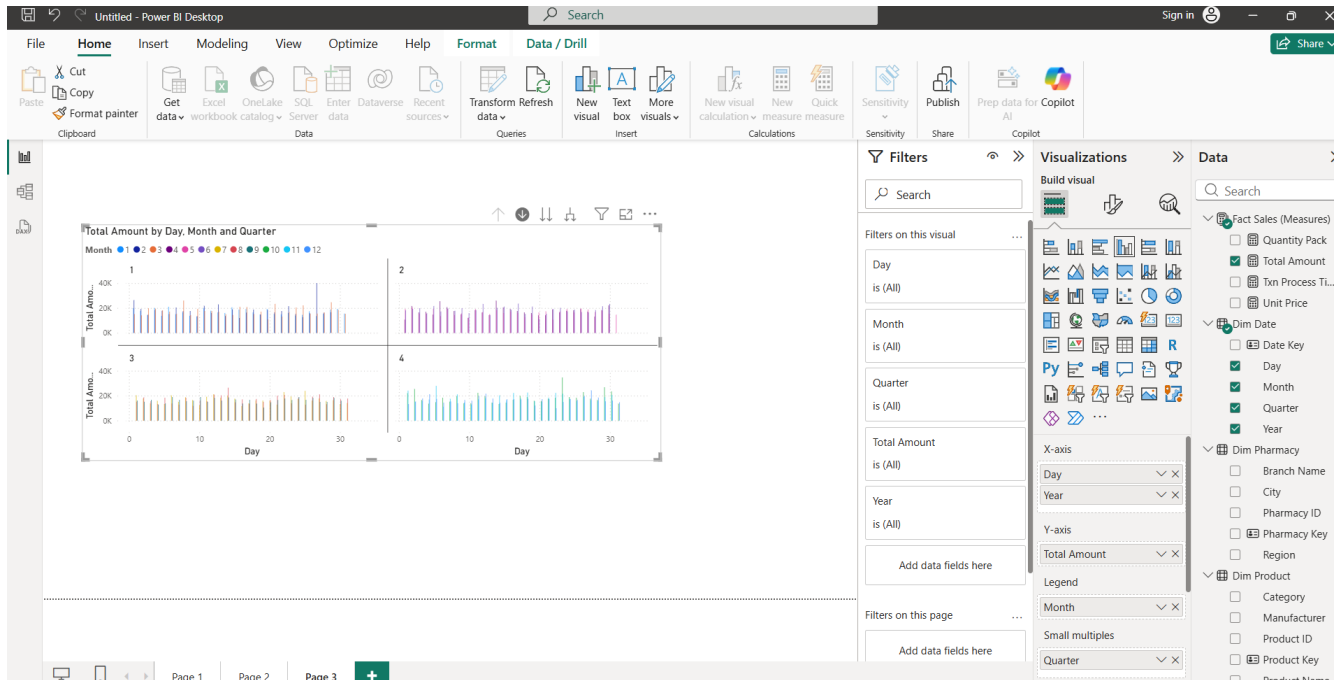
- Added two separate Slicer visuals. I assigned City to the first slicer and Branch Name to the second.



3. Report 3

- Created a Clustered Column Chart with the Date Hierarchy on the X-axis and Total Amount on the Y-axis.





4. Report 4

- In the Drill-through settings of the destination page, I added the Branch Name field as the filter.

The screenshot shows the Power BI Desktop interface with a table of data. A context menu is open over the table, showing options like Copy, Expand, Collapse, Show as a table, Include, Exclude, Drill through, Group, Clear selections, Summarize, Unfreeze row headers, and Set up a verified answer. The 'Drill through' option is selected, and a 'Branch Details' page is being created. The 'Filters' pane on the right shows filters for Branch Name, Quarter, Total Amount, and Year. The 'Visualizations' pane shows a bar chart visualization. The 'Data' pane shows the data model with Fact Sales (Measures), Dim Date, Dim Pharmacy, and Dim Product tables.

Branch Name	1	2	3
Website	610,366.87	431,830.74	
Main St			
Downtown			
Total			

The screenshot shows the Power BI Desktop interface with a table of data. The 'Drill through' settings are configured for the 'Branch Details' page. The 'Filters on this page' pane shows filters for Branch Name, Quarter, Total Amount, and Year. The 'Visualizations' pane shows a bar chart visualization. The 'Data' pane shows the data model with Fact Sales (Measures), Dim Date, Dim Pharmacy, and Dim Product tables.

Product Name	Quantity Pack	Unit Price	Total Amount
2500	1	96.96	96.96
3 days show slimming	1	14.14	14.14
3000	7	386.76	676.83
3m earfit	1	68.25	68.25
4000	1	19.13	19.13
4500	1	42.89	42.89
5 cereals , 5 fruits & milk (6+)	2	367.68	367.68
5000	11	273.35	601.37
5m 1 beauty care massager	1	177.41	177.41
A.M-E.C	145	22,900.68	25,156.05
absolut torr	13	619.20	670.80
A/C Cleanser Sal-Wash	2	306.52	306.52
Accu-Check 50Next Strips	14	475.08	554.26
Aciclovir labetalol 10g	2	128.26	128.26
A-Clavsol 1000mg	3	222.84	222.84
ACM DEPIWHITE ADVANCED	58	710.50	710.50
Acne patch	1	129.81	129.81
acne-clear invisible dots	1	17.46	17.46
Total	20911	1,863,025.72	2,075,661.15

